Anisotropic flow analysis with two- and multiparticle correlation techniques

March 11, 2026

Question 1: *Plot the long $2\theta/\omega$ scan of sample B in a logarithmic scale. Which peaks can you identify?*

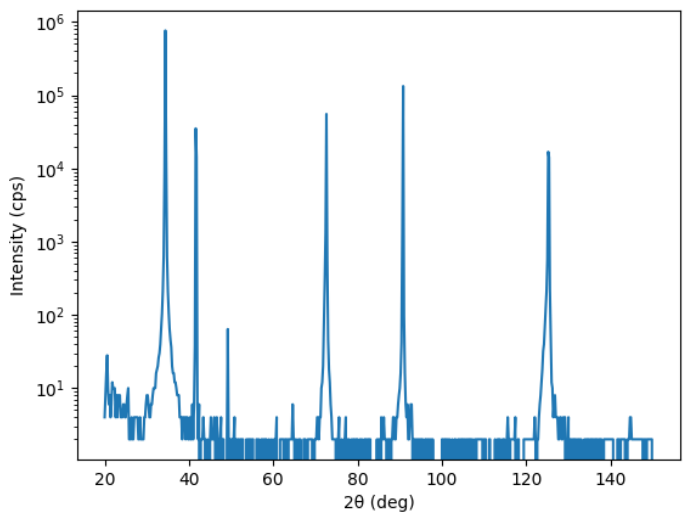

Figure 1: $2\theta/\omega$ wide range scan of sample B

The peaks correspond to Bragg reflections of the ZnO film and the Al_2O_3 substrate.

- $2\theta = 34.5° \rightarrow ZnO$ (002)
- $2\theta = 41.7° \rightarrow Al_2O_3$ (006)
- $2\theta = 72.6° \rightarrow ZnO$ (004)
- $2\theta = 90.8° \rightarrow Al_2O_3$ (0012)
- $2\theta = 125.3° \rightarrow ZnO$ (006)

Question 2: *Estimate the layer thickness of sample A by using Equation (41).*

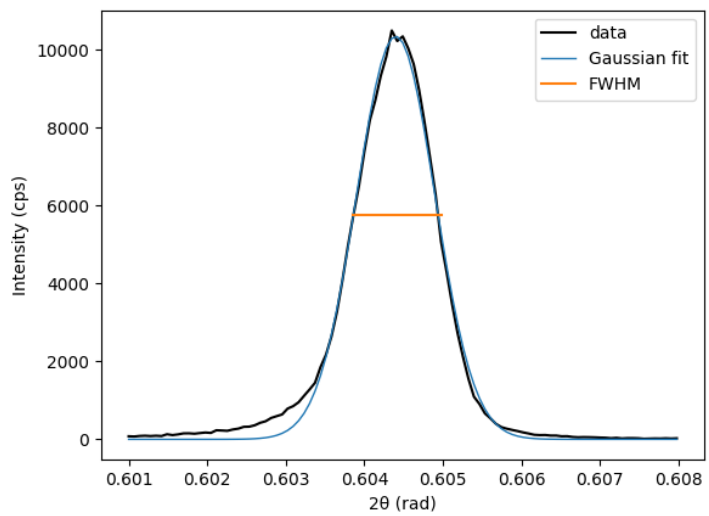

Figure 2: $2\theta/\omega$ scan of sample A and Gaussian fit

According to equation (41):

$$l_z = \frac{0.9\lambda}{\Delta(2\theta)\cos\theta} \tag{1}$$

and $FWHM \approx 0.0011$ from Gaussian fit, layer thickness lz is estimated as 130 nm.

Question 3: *Determine the layer thickness of sample B by using Equations (41) and (42). Compare the obtained values! Conclusion?*

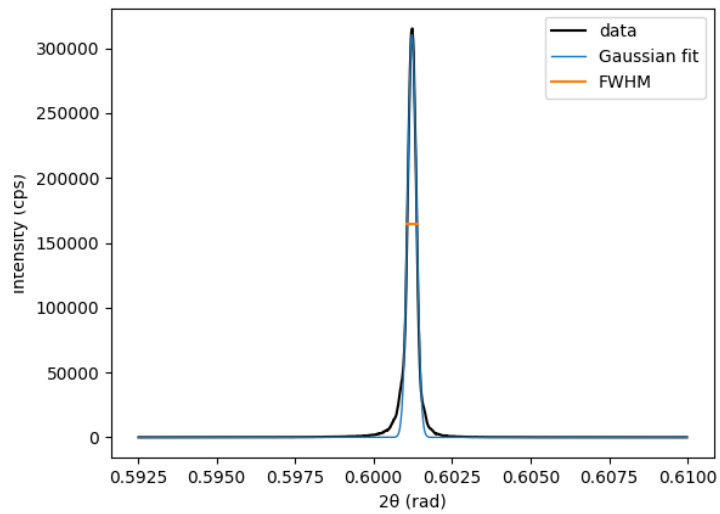

Figure 3: $2\theta/\omega$ scan of sample B and Gaussian fit

According to equation (41) (equation (1) in this report) and $FWHM \approx 0.0003$ from Gaussian fit, layer thickness lz is estimated as 452 nm. Equation (42):

$$l_z = \frac{\lambda \sin\theta}{\Delta(\theta)\sin 2\theta} \tag{2}$$

requires the distance between secondary maxima. However, in the measured $2\theta/\omega$ scan of sample B, only a single peak is visible and no clear secondary maxima can be identified. Therefore, equation (42) cannot be applied to this dataset.

This suggests that the thickness estimate from equation (41) is the only accessible one for sample B from the present measurement.

Question 4: *Estimate the screw-type dislocation densities of samples A and B and their lateral crystallite size. Compare the structural quality of both samples.* By assuming that bandwidth of rocking curve is only caused by limited size, lower limit for the lateral crystallite size $l_{\|}$ can be estimated by:

$$l_{\|} = \frac{0.9\lambda}{\Delta\omega \sin\theta} \tag{3}$$

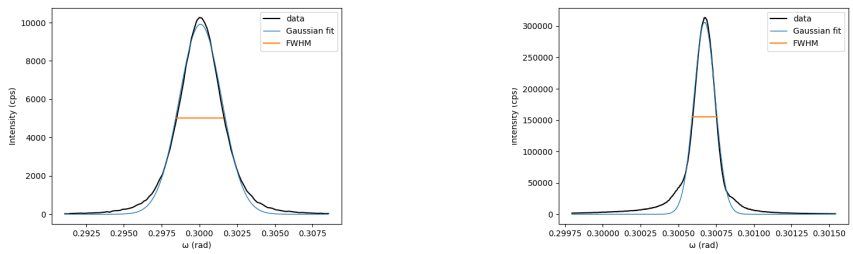

Figure 4: ω scan of sample A and B and Gaussian fit

With FWHM 0.003 for sample A and 0.00016 for sample B, lateral sizes $l_{\|}$ are 149 nm and 2854 nm for sample A and B respectively.

Equation (45) suggests the relation between screw-type dislocation density ρ_{screw} and FWHM $\Delta\omega_{002}$:

$$\rho_{screw} = \frac{\Delta\omega_{002}^2}{4.35|\mathbf{b}_s crew|^2} \tag{4}$$

with the Burger vector $\mathbf{b}_{screw} = [001]$ which equals to lattice constance $c = 5.2024\mathring{A}$. ρ_{screw} are $8.38 \times 10^{12} m^{-2}$ and $2.29 \times 10^{10} m^{-2}$.

Therefore, sample B has a much larger lateral crystallite size and a much lower screw-type dislocation density than sample A, indicating a better structural quality.